

IT3708 - Bio-Inspired Artificial Intelligence

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Examples of Exam Questions

Question 1

☐ 1. Multiple Choice: Among the following options, which op...

Points: 1

Question	Among the following options, which option is not considered a "pillar of evolution"?
Answer	Population Diversity Heredity Constraints Selection

Question 2

☐ 2. Multiple Choice: Among the following operations, which...

Points: 1

Question	Among the following operations, which is not part of an evolutionary algorithm's generation cycle?
Answer	Mutation Evaluation Initialization Crossover Selection Reproduction

Question 3

☐ 3. Multiple Choice: In which type of genetic algorithm (G...

Points: 1

Question	In which type of genetic algorithm (GA) is one individual generated per generation, replacing one individual in the population?
Answer	Crowding GA Generational GA Steady-state GA Messy GA Elitist GA

Question 4

☐ 4. Multiple Choice: What is a potential unfortunate conse...

Points: 1

Question	What is a potential unfortunate consequence of too high selection pressure in an evolutionary algorithm that is performing optimization?
Answer	The population converges to individuals at or close to a local optimum that is not a global optimum. The fitness of the population does not increase much. All individuals in the population converge to be exactly alike, at a local optimum that is not a global optimum. All individuals in the population converge to be exactly alike, at a global optimum. The population converges to individuals at or close to a global optimum.

Question 5

☐ 5. Multiple Choice: Which of the following is not considered...

Points: 1

Question	Which of the following is not considered a space in which to preserve diversity in an evolutionary algorithm?
Answer	Algorithm space Probability space Genotype space Phenotype space

Question 6

☐ 6. Multiple Choice: Which of the following is not considered...

Points: 1

Question	Which of the following is not considered a diversity-preserving genetic algorithm?
Answer	Deterministic crowding genetic algorithm Fitness sharing genetic algorithm Probabilistic crowding genetic algorithm Simple genetic algorithm Island model parallel genetic algorithm

Question 7

7. Multiple Choice: Local search or stochastic local sear...

Points: 2

Question	Local search or stochastic local search methods are often used for hybridising evolutionary algorithms (EAs). In which of the following EA components can, in principle, local search or stochastic local search methods be integrated?
Answer	For initialization only. For initialization or survivor selection only. For initialization, after parent selection, after mutation, or after crossover. For initialization or after parent selection, but never after mutation. For initialization or after parent selection, but never after crossover.

Question 8

8. Multiple Choice: Parameter setting is important in evo...

Points: 1

Question	Parameter setting is important in evolutionary algorithms, and there are different methods for setting parameters. Consider the following description of parameter setting: "a feedback rule based on a measure that monitors search progress." Which type of parameter setting method is this?
Answer	Parameter tuning Deterministic parameter control Adaptive parameter control Self-adaptive parameter control

Question 9

9. Multiple Choice: Parameter setting is important in evo...

Points: 1

Question	Parameter setting is important in evolutionary algorithms, and there are different methods for setting parameters. Consider the following description of parameter setting: "parameter values evolve along with individuals; the values are encoded in the chromosomes and therefore they undergo variation and selection along with the rest of the chromosome." Which type of parameter setting method is this?
Answer	Parameter tuning Deterministic parameter control Adaptive parameter control Self-adaptive parameter control

Question 10

10. Multiple Choice: Crowding methods are well-established...

Points: 2

Question	Crowding methods are well-established methods for preserving diversity in evolutionary algorithms (EAs). Which of the following statements about crowding is not true?
Answer	The competition phase of an evolutionary algorithm is where crowding differs the most from traditional EAs. In generalized crowding, the trade-off between exploration and exploitation can be varied by using a scaling factor parameter. When the scaling parameter is 0, generalized crowding equals deterministic crowding. When the scaling parameter is 2, generalized crowding equals probabilistic crowding. Adaptive generalized crowding employs a method for parameter control.

Question 11

11. Multiple Choice: This problem is concerned with the mi...

Points: 1

Question	This problem is concerned with the minimization of two fitness functions f_1 and f_2. In particular, consider the points $\{x_1, x_2, x_3, x_4\}$ and their fitness function values for a multiobjective minimization problem: • $f_1(x_1) = 1, f_2(x_1) = 1,$ • $f_1(x_2) = 1, f_2(x_2) = 2,$ • $f_1(x_3) = 2, f_2(x_3) = 1,$ and • $f_1(x_4) = 2, f_2(x_4) = 2.$ Which point dominates all the others?
Answer	x1 x2 x3 x4 None

Question 12

12. Multiple Choice: This problem is concerned with the mi...

Points: 1

Question

This problem is concerned with the minimization of two fitness functions f_1 and f_2 .

In particular, consider the points $\{x_1, x_2, x_3, x_4\}$ and their fitness function values for a multiobjective minimization problem:

- $f_1(x_1) = 1, f_2(x_1) = 1,$
- $f_1(x_2) = 1, f_2(x_2) = 2,$
- $f_1(x_3) = 2, f_2(x_3) = 1,$ and
- $f_1(x_4) = 2, f_2(x_4) = 2.$

Which point is nondominated?

Answer

x1

x2

x3

x4

None